Cloud

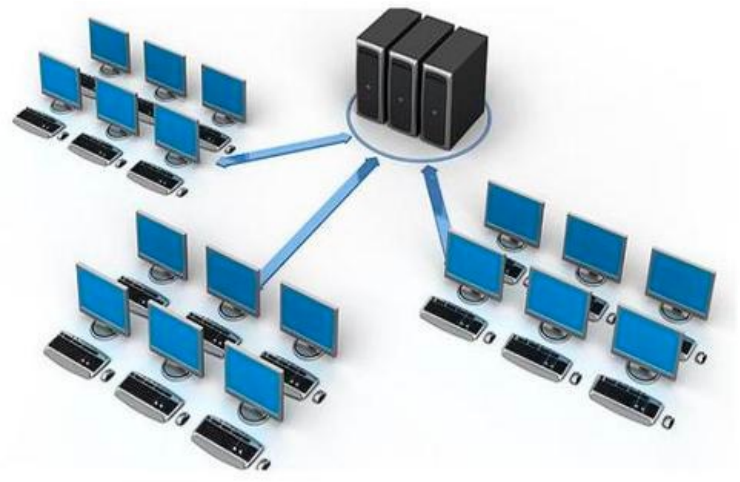
Network

Virtualization

Network Primer

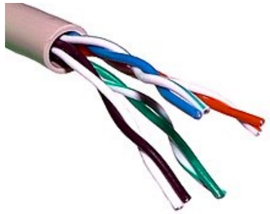
What is network ?

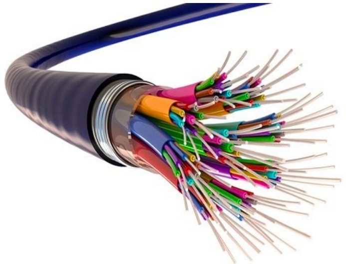
- It is the interconnection of multiple devices, generally termed as Hosts connected using multiple paths
- The purpose of network is: sending/receiving data or media
- It involves various devices like hubs, switches, routers etc.



Wired network

- The network build by connecting devices together using wires/cables as a medium to transfer the data
- Cables
  - Coaxial cable
  - Twisted pairs cables
  - Fiber optics





Wireless network

- The network build by connecting devices together using air as a medium to transfer the data
- EM Waves are used to transfer data from sender to receiver

Network Types

- Personal Area Network
  - Smallest network which is very personal to the user
  - E.g. BlueTooth
- Local Area Network
  - Spans across building(s) and operated under single administrative system
  - E.g. company, school network
  - Technologies: TokenRing or Ethernet
- Metropolitan Area Network
  - Spans across cities
  - E.g. cable network
  - Technologies: high speed fiber optics
- Wide Area Network
  - Spans across countries
  - Technologies: ATM, Frame Relay

What is a network topology ?

- Physical arrangement of computers is known as topology
- Famous topologies
  - Bus
  - Ring
  - Token Ring
  - Star
  - Mesh

ISO OSI model

- Conceptual model that characterizes and standardizes the communication functions of a telecommunication or computing system without regard to its underlying internal structure and technology
- Goal is the interoperability of diverse communication systems with standard communication protocols
- Layered architecture having 7 layers
    - Application
    - Presentation
    - Session
    - Transport
    - Network
    - Data Link
    - Physical

Application Layer

- Specifies interface methods used by hosts in a communications network
- Contains communication protocols
  - **HTTP [80]**: Hyper Text Transfer Protocol
  - **HTTPs [443]**: Secure Hyper Text Transfer Protocol
  - **FTP [20, 21]**: File Transfer Protocol
  - **SFTP [115]**: Simple FTP
  - **DNS [53]**: Domain Name Service
  - **NFS [1023]**: Network File System
  - **POP3 [110]**: Post Office Protocol
  - **SMTP [25]**: Simple Mail Transfer Protocol
  - **SSH [22]**: Secure Shell
  - **LDAP [389]**: Lightweight Directory Access Protocol

Presentation Layer

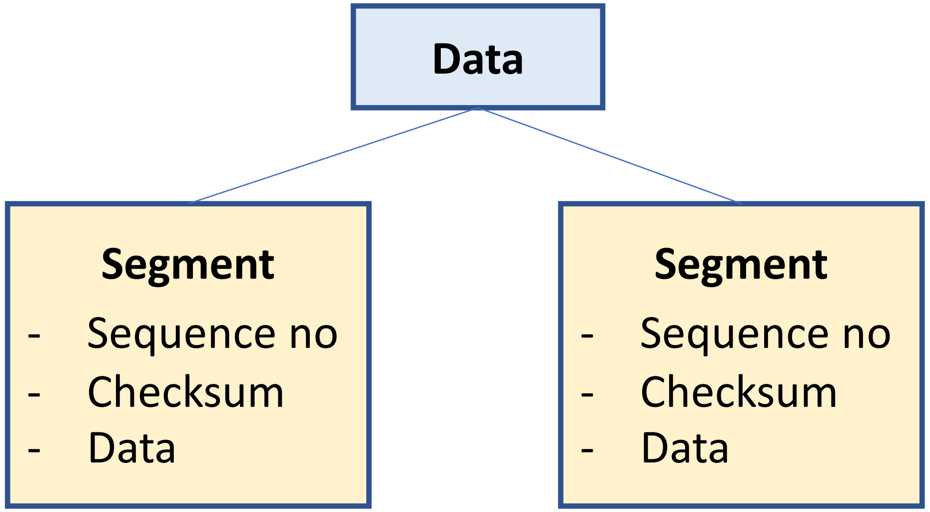
- Serves as the data translator for the network
- Also known as syntax layer
- Responsible for
  - Translation
  - Compression/Decompression
  - Encoding/Decoding
  - Encryption/Decryption

Session Layer

- Provides mechanism for opening, closing and managing session between processes
- Communication sessions consist of requests and responses that occur between applications
- Protocols
  - **ASP**: AppleTalk Session Protocol
  - **ADSP**: AppleTalk Data Stream Protocol
  - **NetBIOS**: Network BIOS
  - **PAP**: Password Authentication Protocol
  - **PPTP**: Point to Point Tunnelling Protocol
  - **RPC**: Remote Procedure Call
  - **SCP**: Session Control Protocol
  - **SDP**: Socket Direct Protocol

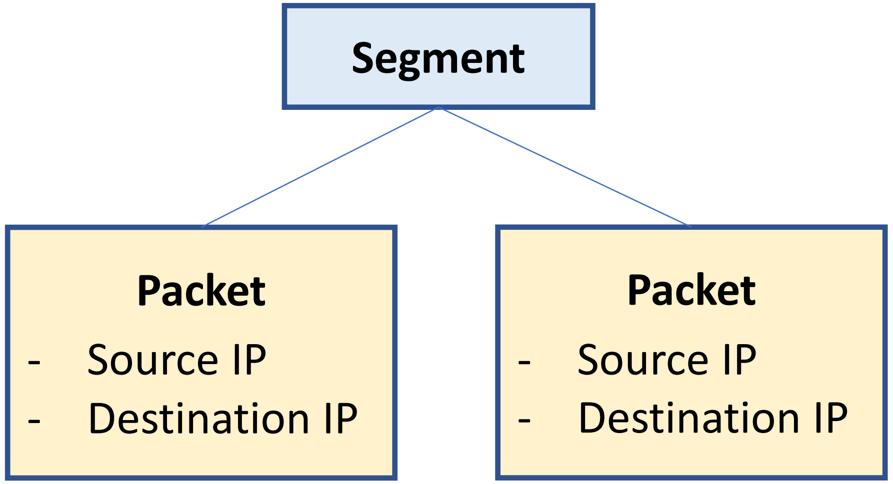
Transport Layer

- Provide host-to-host communication services for applications
- Creates Segment (data unit) containing
  - Sequence number
  - Checksum
  - Port number
- Protocols
  - TCP
    - Connection oriented protocol
    - Provides: Flow Control, Error checking
    - Guarantees data delivery
    - Slower than UDP
    - E.g. WWW, HTTP
  - UDP
    - Connectionless protocol
    - Does not provide flow control
    - Does not guarantee data delivery
    - Faster than TCP
    - E.g. streaming, online games



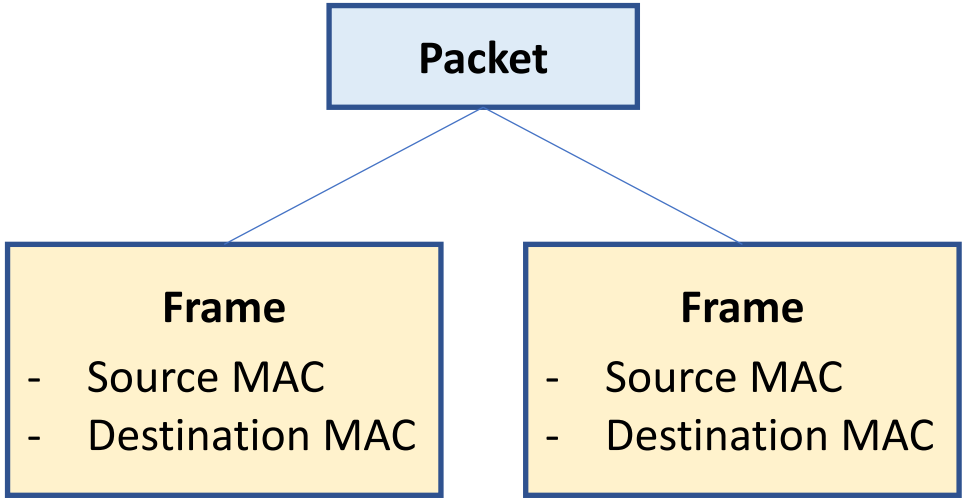
Network Layer

- Responsible for packet forwarding including routing through intermediate routers
- Responsible for splitting segment into packets containing
  - Source IP address
  - Destination IP address
- Protocols
  - **IP**: Internet Protocol
  - **IPX**: Internetwork Packet Exchange
  - **IPSec**: Internet Protocol Security
  - **EGP**: Exterior Gateway Protocol



Data Link Layer

- Transfers data between
  - adjacent network nodes in a wide area network (WAN) or
  - between nodes on the same local area network (LAN) segment
- Encapsulates packet into Frames containing
  - Source MAC Address
  - Destination MAC Address
- Sublayers
  - Logical Link Layer
  - Media Access Control Layer



Data Link Layer: Logical Link Layer

- The uppermost sublayer multiplexes protocols running at the top of data link layer, and optionally provides flow control, acknowledgment, and error notification
- Provides addressing and control of the data link
- Services
  - Error control (automatic repeat request, ARQ)
  - Flow control [Data-link-layer flow control is not used in LAN protocols such as Ethernet, but in modems and wireless networks]

Data Link Layer: Media Access Control Layer

- Refers to the sublayer that determines who is allowed to access the media at any one time (CSMA/CD)
- Determines where one frame of data ends and the next one starts (frame synchronization)
- Frame synchronization uses: time based, character counting, byte stuffing and bit stuffing.
- Services
  - Multiple access protocols for channel-access control,
    - CSMA/CD protocols for collision detection and re-transmission in Ethernet networks
    - CSMA/CA protocol for collision avoidance in wireless networks
  - Physical addressing (MAC addressing)
  - LAN switching (packet switching), including MAC filtering, Spanning Tree Protocol (STP) and Shortest Path Bridging (SPB)
  - Data packet queuing or scheduling

Physical Layer

- Consists of the electronic circuit transmission technologies of a network
- Fundamental layer underlying the higher level functions in a network which provides means of transmitting raw bits rather than logical packets or segments
- The bitstream may be grouped into code words or symbols and converted to a physical signal that is transmitted over a transmission medium
- Translates logical communications requests from the data link layer into hardware-specific operations to cause transmission or reception of electronic signals
- Services
  - Modulation/Demodulation
  - Multiplexing
- Consists of
  - Cables/wires
  - Devices like hub, repeaters etc.

Addressing Modes: MAC Address

- Used to identify NIC uniquely
- Consists of 6 bytes [48 bits]
- First 3 bytes represents manufacturer
- Next 3 bytes represents NIC's unique address

Addressing Modes: IP Address

- Used to identify every device uniquely
- Set by operating system running on the device
- Can be written in
  - Decimal: 192.168.100.10
  - Binary: 11000000.10101000.01100100.00001010
- Types
  - Private: used to communicate with other devices in local network
  - Public: used to communicate with other devices over internet
- Versions
  - IPv4
    - 32 bit [4 bytes] address
    - Classful and Classless addressing
  - IPv6
    - 128 bit address

IP Address: Classful addressing

Class A	Class B	Class C	Class D	Class E
0.x.x.x – 127.x.x.x	128.x.x.x to 191.x.x.x	192.x.x.x to 223.x.x.x	224.x.x.x to 239.x.x.x	240.x.x.x to 255.x.x.x
IP addresses start with: 0	IP addresses start with: 10	IP addresses start with: 110	IP addresses start with: 1110	IP addresses start with: 1111
Private: 10.x.x.x	Private: 172.16.x.x to 172.31.x.x	Private: 192.168.x.x	**Reserved for multicast**	**Reserved for RnD**
Net mask: 255.0.0.0	Net mask: 255.255.0.0	Net mask: 255.255.255.0		

IP Address: Classless addressing

- To reduce the wastage of IP addresses in a block, we use sub-netting
- Dividing a large block of addresses into several contiguous sub-blocks and assigning these sub-blocks to different smaller networks is called subnetting
- Also known as Classless Interdomain Routing (CIDR)
- Uses variable length subnet mask (VLSM)
- E.g. 172.168.100.12/28

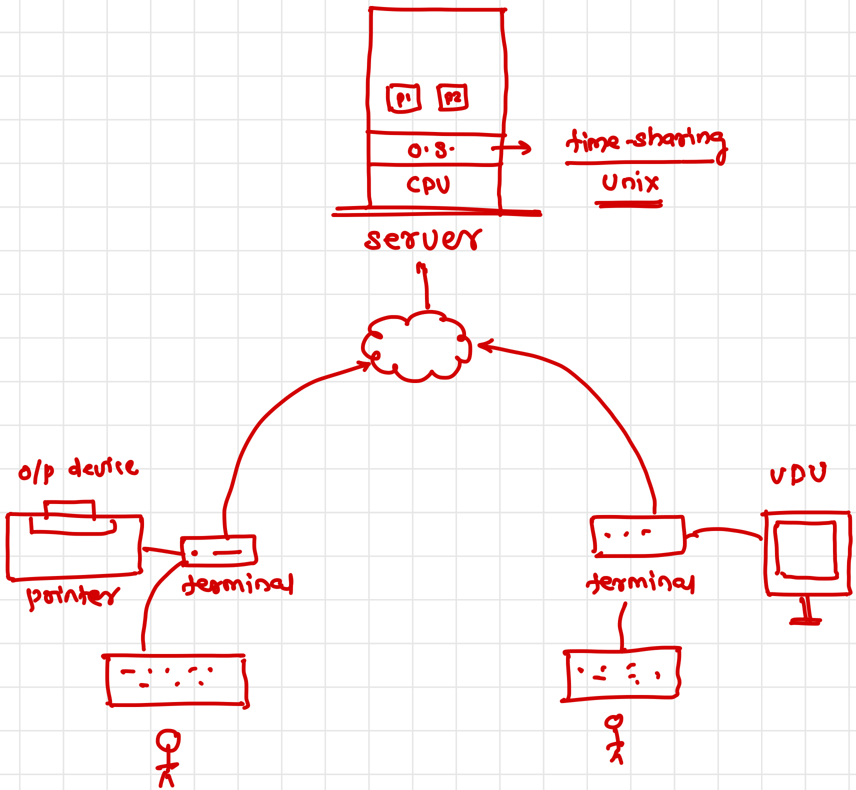
Cloud

Computing Model

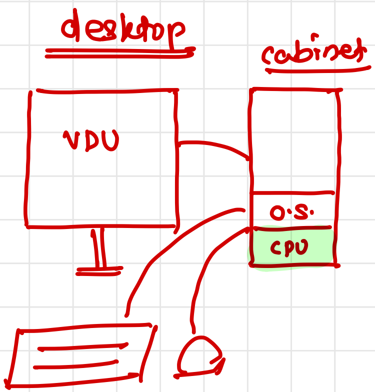
- Desktop computing
- Client-Server computing
- Cluster computing
- Cloud Computing

o] Pre-Desktop

computer
↓
computes → CPU



i) desktop computing



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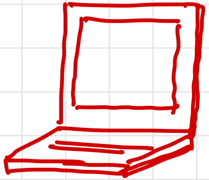
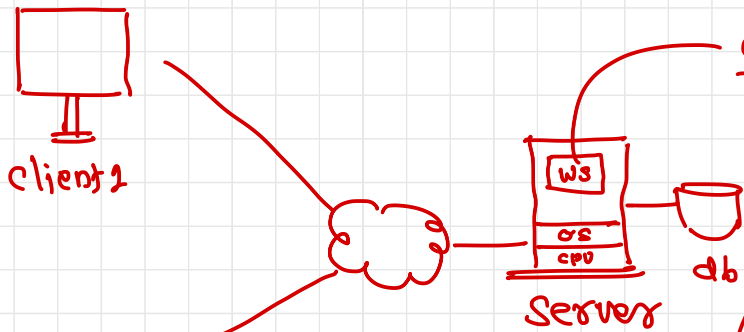


2] Client-server computing

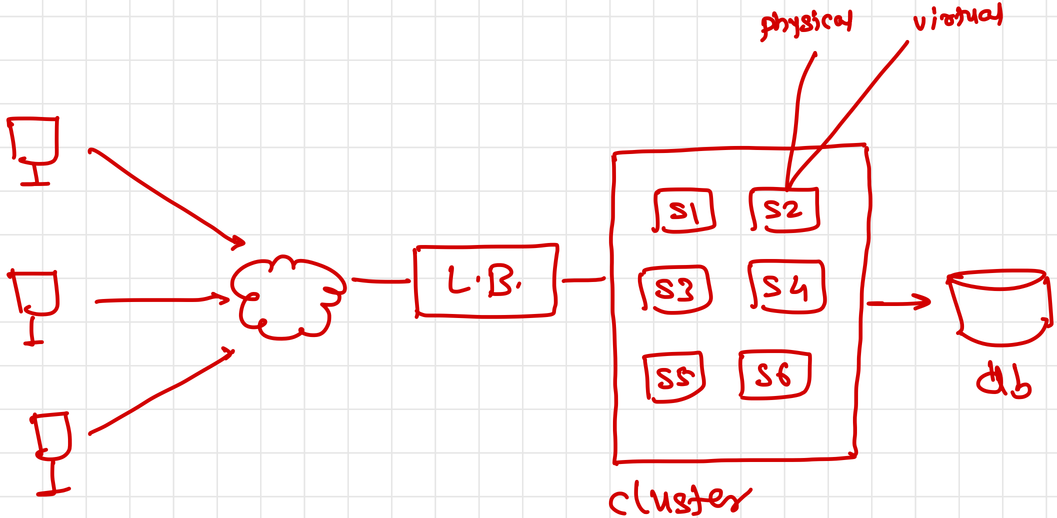


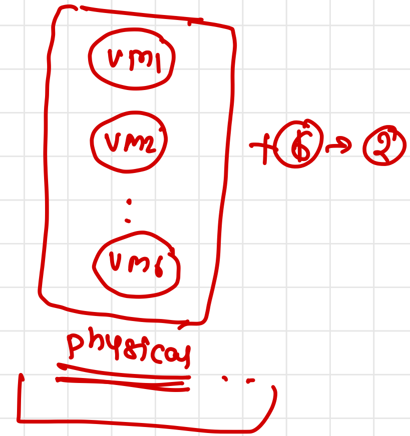
web server
- apache
- nginx

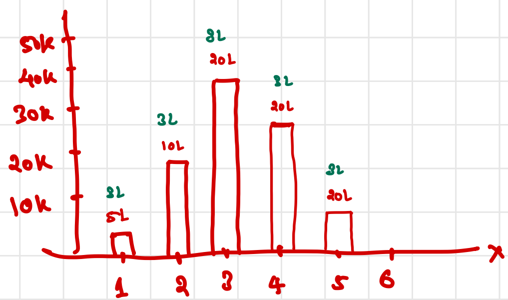
RDBMS
- mysql
- oracle
- ms sql server
- PostgreSQL
- DB2

NoSQL
- mongo DB
- Redis
- cassandra
- Redshift
- neo4J
- ⋮

3) cluster computing





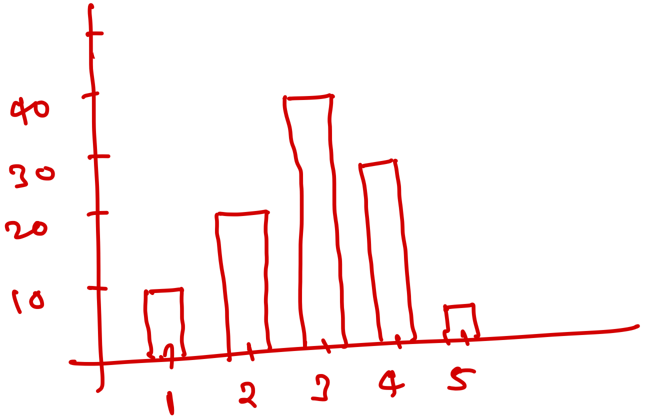


○ Capital Expenditure } CapEx

Operational expenditure } OpEx

What is cloud computing ?

- The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer.
- Is the delivery of on-demand computing resources – everything from data centers over the internet on a pay for use basis
- Cloud computing is an umbrella term used to refer to Internet based development and services



OpeEx

VM → 1$

VM → 2$

+ 2VM → 4$

- 1VM = 3$

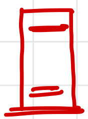
- 2VM = 1$

What is Data Center ?

- Where your IT devices and applications are located
- For a non-technical person it is the cloud where the user's files/data is stored
- Components
  - Servers
  - Security
  - WAN
  - Storage
  - File Sharing

Server

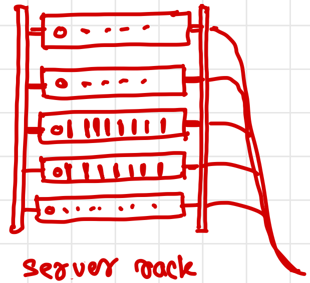
tower



rack

24-CPUs → compute

hard disk → storage → 100 TB



server rack

rack server
↳ rack of servers (20)
↳ rack row (20 racks)
↳ room (20 rows)
↳ floor (10 rooms)
↳ building (5 flrs)
↳ buildings
↳ data Center (3)
↳ availability Zone (2dc)
↳ region
↳ provider

virtual H/W components
Bare metal Hypervisor → VMWare ESXi
HAL
Physical Hardware

What is Virtualization ?

- Refers to the act of creating a virtual (rather than actual) version of something, including virtual computer hardware platforms, storage devices, and computer network resources
- Types
  - Type I
  - Type II
  - Containerization

Terminologies

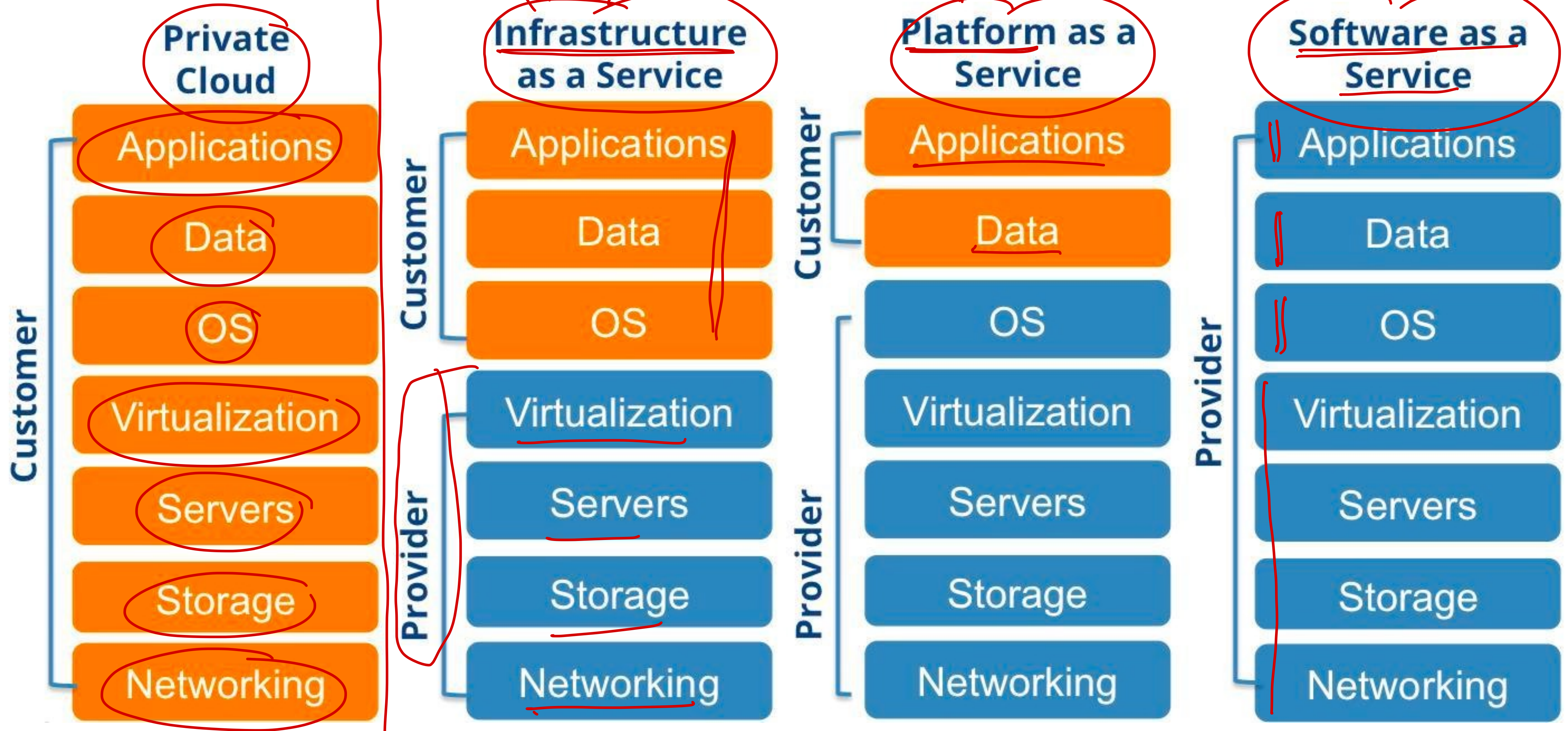
- Scalability
  - refers to the idea of a system in which every application or piece of infrastructure can be expanded to handle increased load
- Elasticity
  - the degree to which a system is able to adapt to workload changes by provisioning and de-provisioning resources in an autonomic manner, such that at each point in time the available resources match the current demand as closely as possible
- Availability
  - refers to the ability of a user to access information or resources in a specified location and in the correct format
- Information Assurance
  - availability, integrity, authentication, confidentiality and nonrepudiation
- On-demand service
  - A model by which a customer can purchase cloud services as needed

Service Models

operations
VM network vpc storage

developers
VM + Environment

end users



Service Models

End Users

Software as a Service (SaaS)

Google Apps, Facebook, YouTube, Dropbox, Google Photos

developers

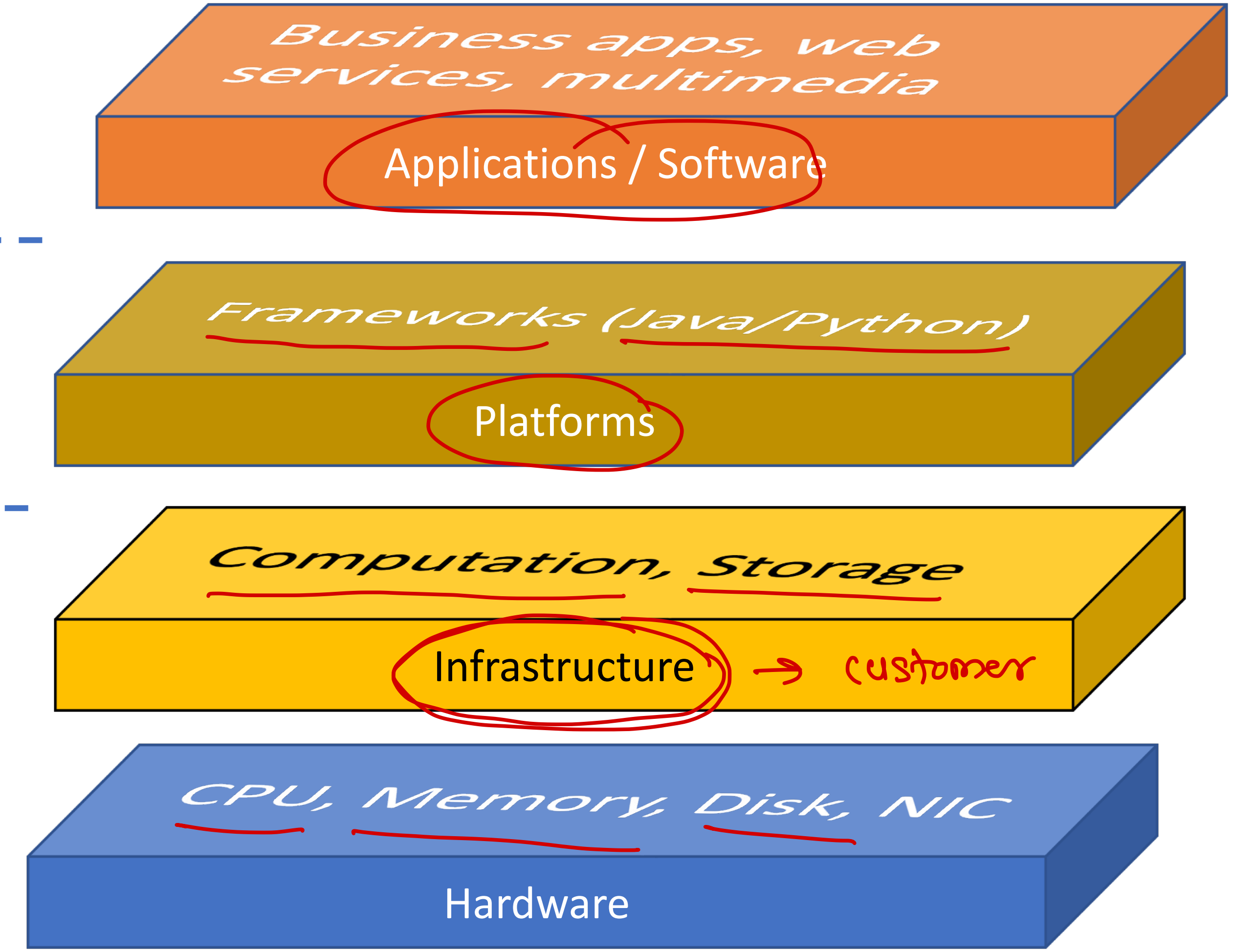
Platform as a Service (PaaS)

Google App Engine, Amazon Simple DB, S3, Microsoft Azure

Operations

Infrastructure as a Service (IaaS)

Amazon EC2, Google Compute VM, Azure VM



Data Center

provider

Models

1. IaaS
2. PaaS
3. SaaS
4. FaaS → function as a service — lambda
5. DaaS → Database -- " -- " — RDS
6. EaaS → Everything — " ——

Service Models: IaaS

- Infrastructure as a Service
- Allocates virtualized computing resources to the user through the internet
- IaaS is completely provisioned and managed over the internet
- helps the users to avoid the cost and complexity of purchasing and managing their own physical servers
- Every resource of IaaS is offered as an individual service component and the users only have to use the particular one they need
- The cloud service provider manages the IaaS infrastructure while the users can concentrate on installing, configuring and managing their software
- Generally meant for operations team to setup the required infrastructure
- Benefits
  - Time and cost savings: more installation and maintenance of IT hardware in-house,
  - Better flexibility: On-demand hardware resources that can be tailored to your needs,
  - Remote access and resource management.

Service Models: PaaS

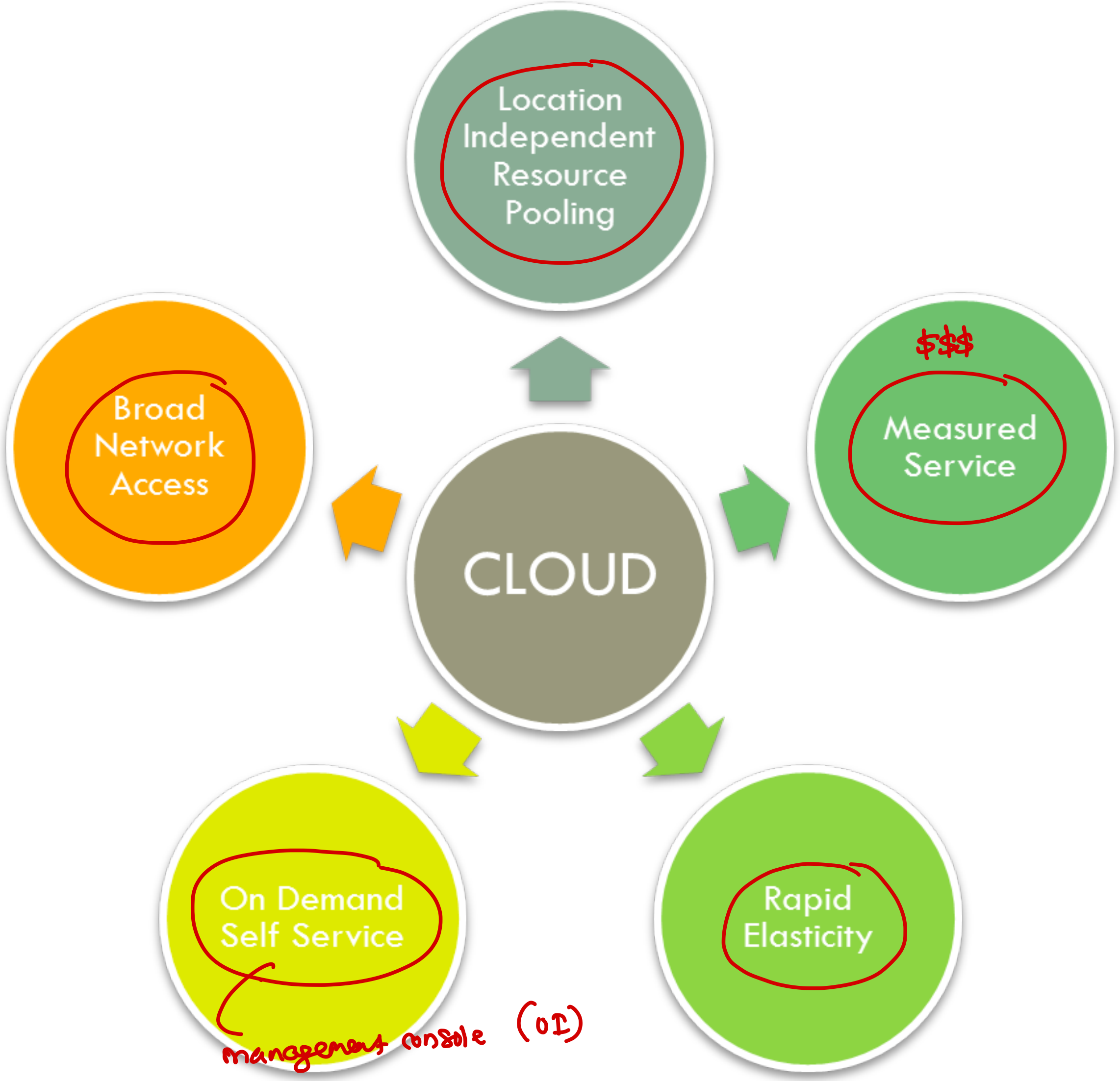
- Provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure typically associated with developing and launching an app
- Generally meant for developers
- Benefits
  - Mastering the installation and development of software applications
  - Time saving and flexibility for development projects: no need to manage the implementation of the platform, instant production
  - Data security: You control the distribution, protection, and backup of your business data

Service Models: SaaS

- Software as a Service
- Software distribution model in which a third-party provider hosts applications and makes them available to customers over the Internet
- User wont know which computer or operating system or infrastructure is used to host the software
- Generally meant for end user
- Benefits
  - You are entirely free from the infrastructure management and aligning software environment: no installation or software maintenance
  - You benefit from automatic updates with the guarantee that all users have the same software version
  - It enables easy and quicker testing of new software solutions.

Cloud Computing Characteristics

- Rapid Elasticity
- On Demand Self Service
- Broad Network Access
- Location Independent Resource Sharing
- Measured Services



Cloud Deployment Models: Public ("internet")

- Supports all users who want to make use of a computing resource, such as hardware (OS, CPU, memory, storage) or software (application server, database) on a subscription basis
- Most common uses of public clouds are for application development and testing, tasks such as file-sharing, and e-mail service
- Requires internet to access the resources

Cloud Deployment Models: Private (security)

- Typically infrastructure used by a single organization
- Such infrastructure may be managed by the organization itself to support various user groups, or it could be managed by a service provider that takes care of it either on-site or off-site
- Private clouds are more expensive than public clouds due to the capital expenditure involved in acquiring and maintaining them
- However, private clouds are better able to address the security and privacy concerns of organizations

Cloud Deployment Models: Hybrid

- Organization makes use of interconnected private and public cloud infrastructure
- Many organizations make use of this model when they need to scale up their IT infrastructure rapidly, such as when leveraging public clouds to supplement the capacity available within a private cloud
- For example, if an online retailer needs more computing resources to run its Web applications during the holiday season it may attain those resources via public clouds.

Cloud Services

- Compute: used to create the Virtual Machine
- Storage: used to provide the storage
- Database: RDBMS + No SQL
- Security and Identity Management
- Media Services
- Machine Learning
- Cost Management
- Application Integration

Advantages

- Lower computer costs
- Improved performance
- Reduced software costs
- Instant software updates
- Improved document format compatibility
- Unlimited storage capacity
- Increased data reliability
- Universal document access
- Latest version availability

Disadvantages

- Requires a constant Internet connection
- Does not work well with low-speed connections
- Features might be limited
- Stored data might not be secure
- Stored data can be lost
- Each cloud systems uses different protocols and different APIs

Cloud Providers

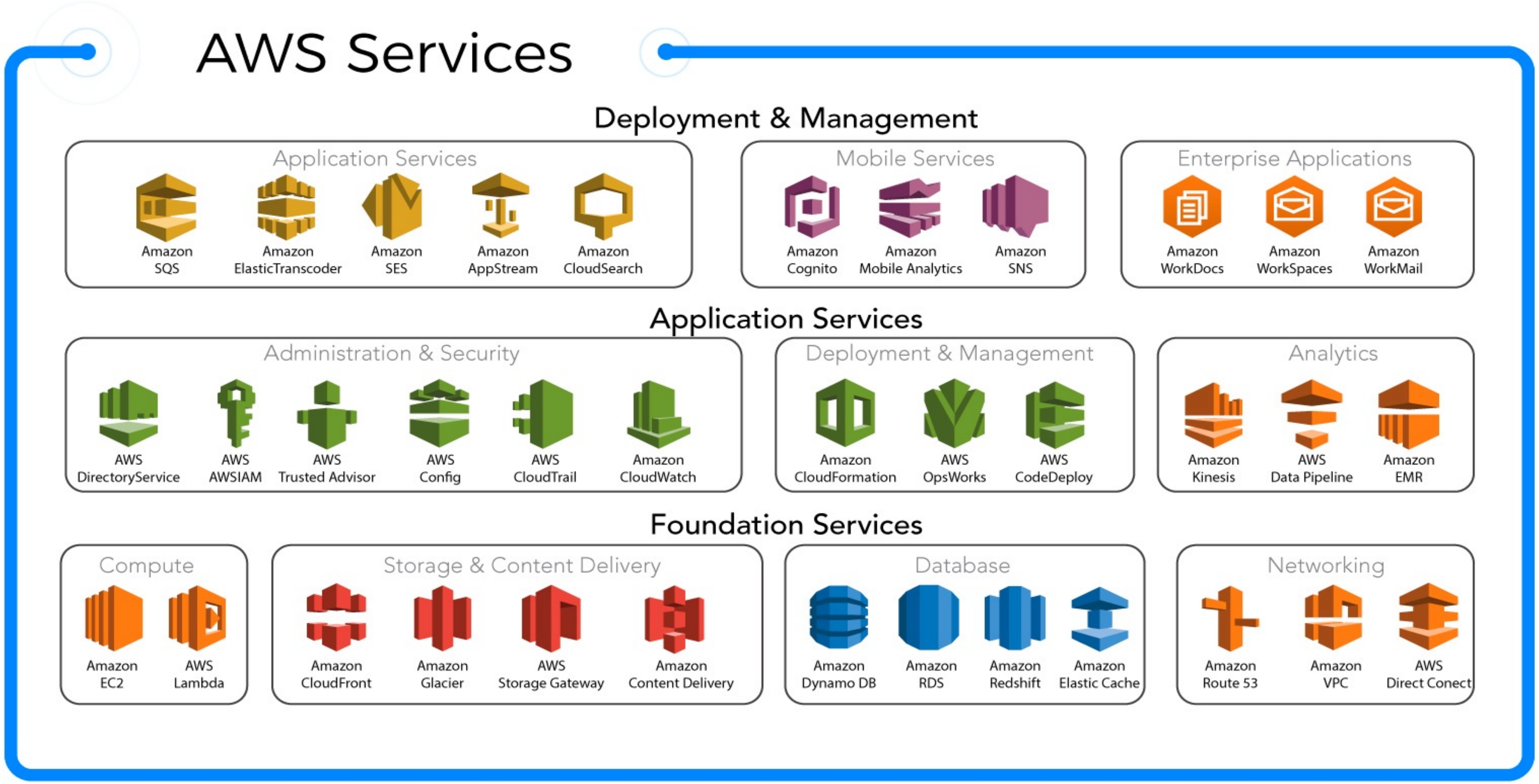
- Amazon Web Services
- Google Cloud Platform
- Microsoft Azure
- Rackspace
- DigitalOcean
- Alibaba Cloud
- Oracle Cloud
- IBM Cloud

AWS

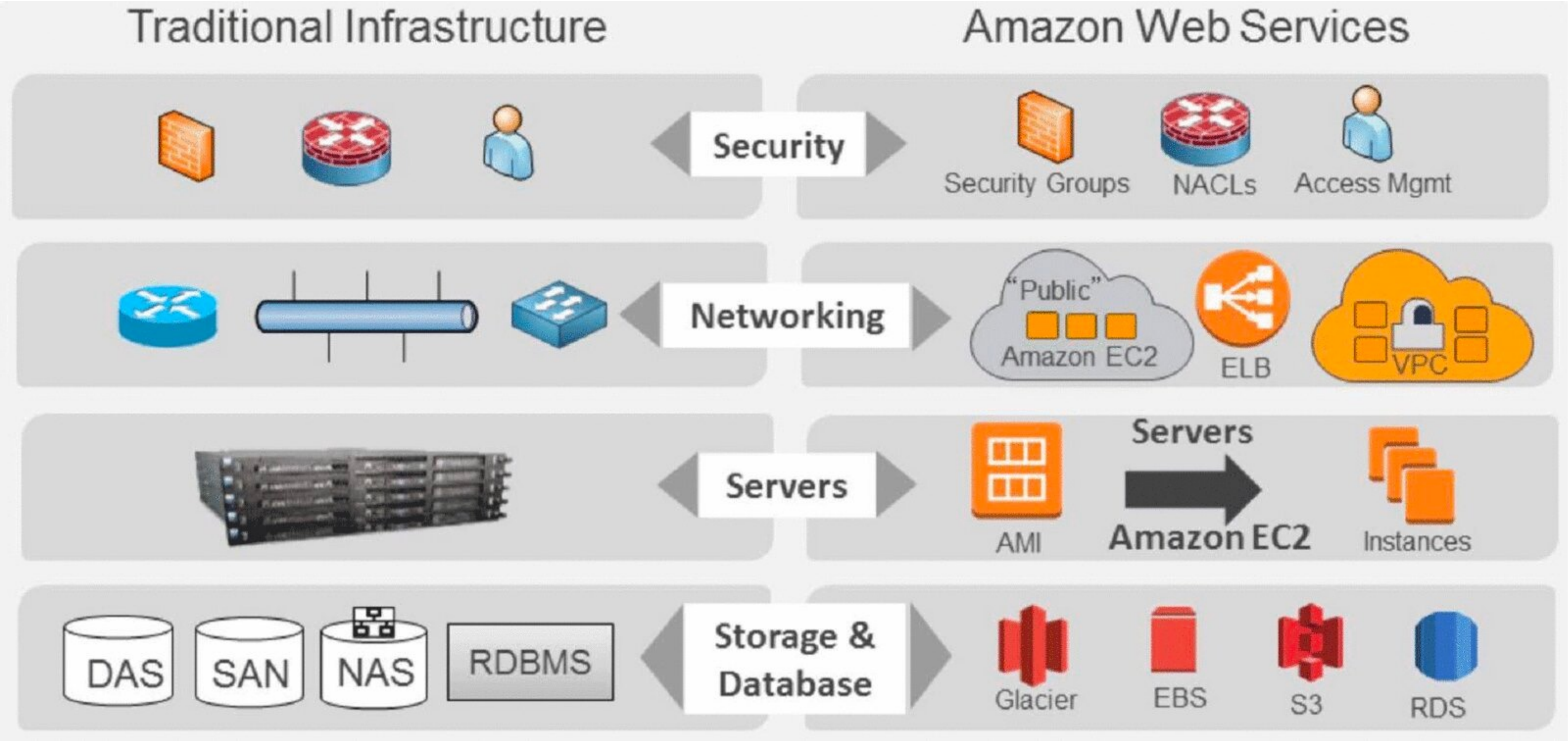
What is AWS ?

- AWS stands for Amazon Web Services
- Platform that offers flexible, reliable, scalable, easy-to-use and cost-effective cloud computing solutions
- Amazon’s cloud implementation
- It’s a combination of IaaS, PaaS and SaaS offerings

AWS Services



Traditional vs AWS



Global Infrastructure

Global Infrastructure: Region

- Geographic area having availability zone(s)
- Collection of availability zones that are geographically located close to one other
- Every Region will act independently of the others, and each will contain at least two Availability Zones
- E.g.
  - US East: N. Virginia, Ohio
  - US West: N. California, Oregon
  - Asia Pacific: Mumbai, Seoul, Singapore, Sydney, Tokyo

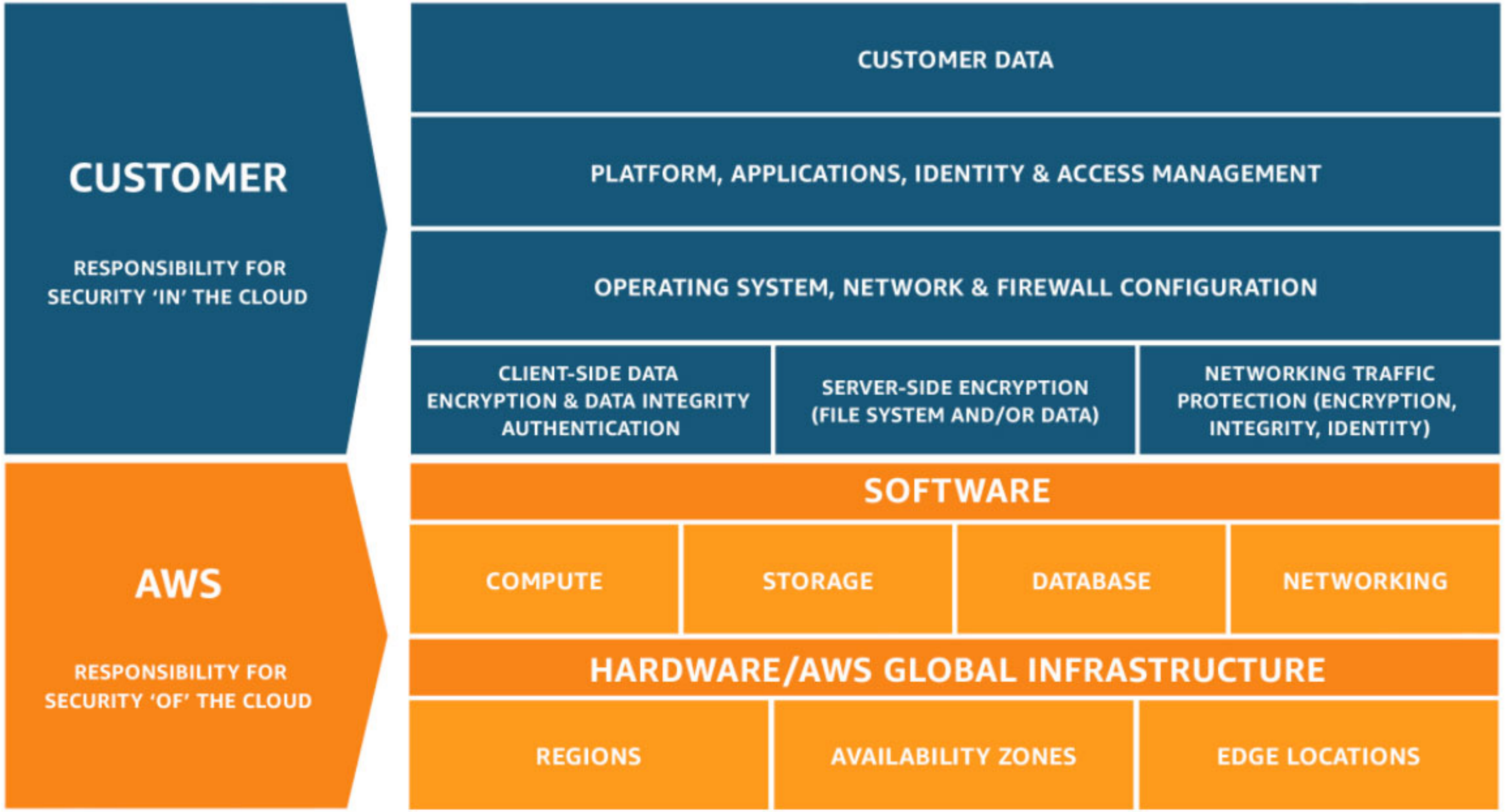
Global Infrastructure: Availability Zone

- Essentially the physical data centers of AWS
- This is where the actual compute, storage, network, and database resources are hosted that we as consumers provision within our Virtual Private Clouds (VPCs)
- Availability Zones are always referenced by their Code Name, which is defined by the AZs Region Code Name that the AZ belongs to, followed by a letter
- E.g.
  - the AZs within the eu-west-1 region (EU Ireland), are
    - eu-west-1a
    - eu-west-1b
    - eu-west-1c

Global Infrastructure: Edge Locations

- Edge Locations are AWS sites deployed in major cities and highly populated areas across the globe
- Generally used to cache data and reduce latency for end-user access by using the Edge Locations as a global Content Delivery Network (CDN)
- Edge Locations are primarily used by end users who are accessing and using your services
- E.g.
  - Route 53: DNS Lookup
  - CloudFront
    - Content Delivery Network (CDN)
    - Cached contents, streaming distribution, acceleration

AWS Shared Responsibility Model



Terminologies

- **High Availability**
  - Refers to architectures that continue to remain available to end users in the event of a component or systems failure
  - On AWS, multi-AZ architectures allow your applications to remain available in the event of AZ outage
- **Fault Tolerance**
  - Refers to the architecture, that not only remain available duration an outage (High Availability) but also suffer no degradation in the performance.
  - Usually requires extra redundancy and should be traded off with cost concerns.
- **Scalability**
  - Refers to the ability of a system to increase in size and capacity in a cost effective way based on demand
  - Scaling can be
    - vertical (increase capacity of a single instance or server)
    - Horizontal (add or terminate number of instances)
- **Elasticity**
  - Refers to the ease of system's ability to change or adapt
  - Like automatically scaling up or down as per user demand

IAM

Introduction

- Used to securely control individual user and group access to AWS resources
- Is a global server and does not apply to any region
- Makes it easy to provide multiple users secure access to AWS resources
- Provides centralized control of your account
- Used to manage
  - Users
  - Groups
  - Access policies
  - Roles
  - User credentials
  - User password policies
  - Multi-factor authentication (MFA)
  - API keys for programmatic access (CLI)

Introduction

- Best practices
  - Do not use the root account for anything other than billing
  - Delete the API keys for root account
  - Protect root account with MFA
  - Always create individual IAM users with required permissions
  - Apply strong IAM password policy
  - Always rotate access keys

Policies

- Policies are the documents that define permissions and can be applied to users, groups and roles
- Policies don’t do anything on its own, it will be used when attached to an entity
- Written in JSON
- With multiple policies the most restrictive policy is applied
- If a request is not explicitly allowed, by default it is implicitly denied
- It a request is explicitly denied, it override everything else.
- The condition element can be used to apply further conditional logic
- Only attached policies have any impact
- When evaluating policies, all applicable policies are merged

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Deny",
            "Action": [
                "s3:Get*",
                "s3:List*"
            ],
            "Resource": "*"
        }
    ]
}
```

Policies

- Can be assigned at
  - IAM Role level
  - IAM Users level
  - IAM Group level
- Policy Types
  - Inline policy
  - Managed policy
    - AWS managed
    - Customer managed

- Always remember DENY > ALLOW > DENY

IAM User

- IAM user is an entity that represents an individual person or service
- “root user" is the account created when you setup the AWS account
- The root account has full administrative permissions and these cannot be restricted
- By default users cannot access anything in your account
- IAM user can have
  - An access key Id and secrete access key for programmatic access to AWS API, CLI and SDK
  - A password to access management console
- You can create up to 5000 users per AWS account (hard limit)
- 10 group membership per IAM user
- Default maximum 10 managed policies per user
- No inline limit but you can not exceed 2048 characters for all inline policies on user
- 1 MFA per user
- 2 access keys per user

ARN

- Amazon Resource Name
- Used to identify every resource uniquely within AWS
- ARNs always begin with:

arn:partition:service:region:account-id

  - Where
    - partition = aws or aws-cn (for china)
    - service = the AWS service: s3, ec2
    - region = region code: us-west-1

- And depending on service, finish with:
  - resource
  - resourcetype/resource
  - resourcetype/resource/qualifier
  - resourcetype/resource:qualifier
  - resourcetype:resource
  - Resourcetype:resource:qualifier

IAM User: Keys

- The Access Key ID and Secret Access Key are not the same as a password and cannot be used to login to the AWS console
- The Access Key ID and Secret Access Key can only be generated once and must be regenerated if lost

IAM Group

- Collection of users is a group
- A group may have policies attached to control the access
- A group is not an identity and cannot be identified as a principal in an IAM policy
- Use groups to assign permissions to multiple users at a time
- Group can NOT be nested

IAM Role

- Role can be used in a scenario where no permanent credentials are required for an entity but needs to grant an access to certain services
- Roles are created and then "assumed" by trusted entities and define a set of permissions for making AWS service requests
- IAM users or AWS services can assume a role to obtain temporary security credentials that can be used to make AWS API calls
- There are no credentials associated with a role (password or access keys)
- A role can be assigned to a federated user who signs in using an external identity provider
- Temporary credentials are primarily used with IAM roles and automatically expire

Setup AWS CLI

- macOS
  - Install homebrew
    - /usr/bin/ruby -e "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/master/install)"
  - Install aws cli
    - brew install awscli
- Windows
  - Visit: https://aws.amazon.com/cli/
  - Download and install installer

- Linux
  - CentOS
    - sudo yum install epel-release
    - sudo yum install python-pip
    - sudo pip install awscli

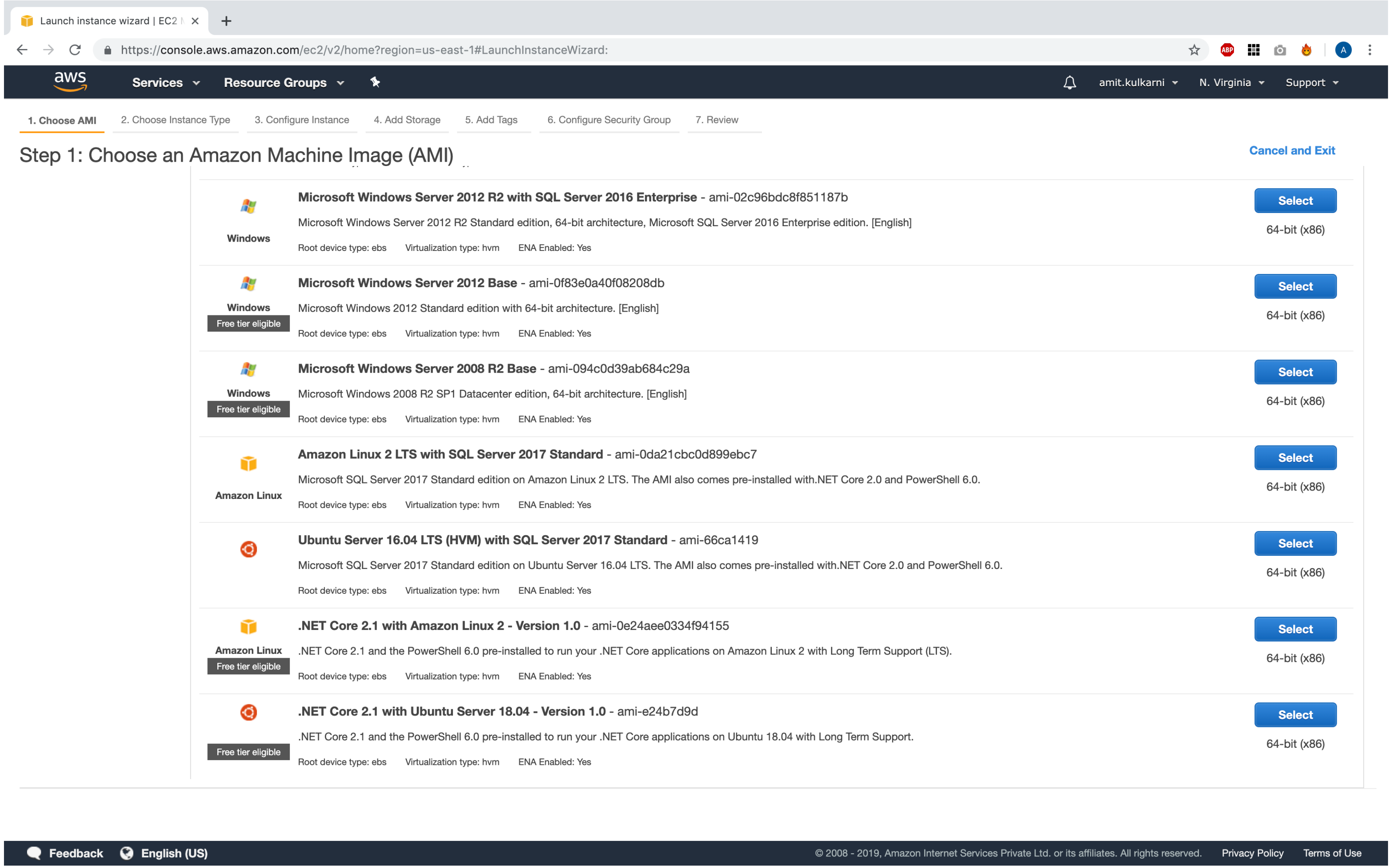
- Linux
  - Ubuntu
    - sudo apt-get install python
    - sudo apt-get install python-pip
    - sudo pip install awscli

EC2

EC2

- Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides resizable compute capacity in the cloud
- It is a virtual machine you will be building in the cloud
- EC2 instances are designed to mimic traditional on-premise servers, but with the ability to be commissioned and decommissioned on-demand for easy scalability and elasticity
- EC2 supports variety of operating systems:
  - Linux: Amazon Linux, Ubuntu, Red Hat Enterprise, SUSE Linux Enterprise Server, Fedora, Debian, CentOS, Gentoo Linux, Oracle Linux, FreeBSD
  - Windows: Windows Server, Windows
- Every instance comprised of
  - Amazon Machine Image (AMI)
  - Instance type
  - Network Interface
  - Storage

Step 1 - Choose AMI



Amazon Machine Image (AMI)

- Operating System used to create virtual machine (EC2 instance)
- AMI are built for a specific region
- You can copy an AMI from one region to another
- You can also create a custom AMI with required applications/configuration
- AMI contains
  - Template for root volume
  - Launch permissions that control which account can use the AMI
  - EBS mapping that specifies the volume(s) to attach the instance when its launched
- AMI comes into two types
  - Instance store backed AMI
  - EBS backed AMI

Step 2 - Instance Type

Launch instance wizard | EC2

https://console.aws.amazon.com/ec2/v2/home?region=us-east-1#LaunchInstanceWizard:

aws Services Resource Groups amit.kulkarni N. Virginia Support

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 2: Choose an Instance Type

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instances are virtual servers that can run applications. They have varying combinations of CPU, memory, storage, and networking capacity, and give you the flexibility to choose the appropriate mix of resources for your applications. Learn more about instance types and how they can meet your computing needs.

Filter by: All instance types | Current generation | Show/Hide Columns

Currently selected: t2.micro (Variable ECUs, 1 vCPUs, 2.5 GHz, Intel Xeon Family, 1 GiB memory, EBS only)

	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance	IPv6 Support
	General purpose	t2.nano	1	0.5	EBS only	-	Low to Moderate	Yes
☑	General purpose	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate	Yes
	General purpose	t2.small	1	2	EBS only	-	Low to Moderate	Yes
	General purpose	t2.medium	2	4	EBS only	-	Low to Moderate	Yes
	General purpose	t2.large	2	8	EBS only	-	Low to Moderate	Yes
	General purpose	t2.xlarge	4	16	EBS only	-	Moderate	Yes
	General purpose	t2.2xlarge	8	32	EBS only	-	Moderate	Yes
	General purpose	t3.nano	2	0.5	EBS only	Yes	Up to 5 Gigabit	Yes
	General purpose	t3.micro	2	1	EBS only	Yes	Up to 5 Gigabit	Yes
	General purpose	t3.small	2	2	EBS only	Yes	Up to 5 Gigabit	Yes
	General purpose	t3.medium	2	4	EBS only	Yes	Up to 5 Gigabit	Yes
	General purpose	t3.large	2	8	EBS only	Yes	Up to 5 Gigabit	Yes
	General purpose	t3.xlarge	4	16	EBS only	Yes	Up to 5 Gigabit	Yes

Cancel | Previous | Review and Launch | Next: Configure Instance Details

Feedback English (US)

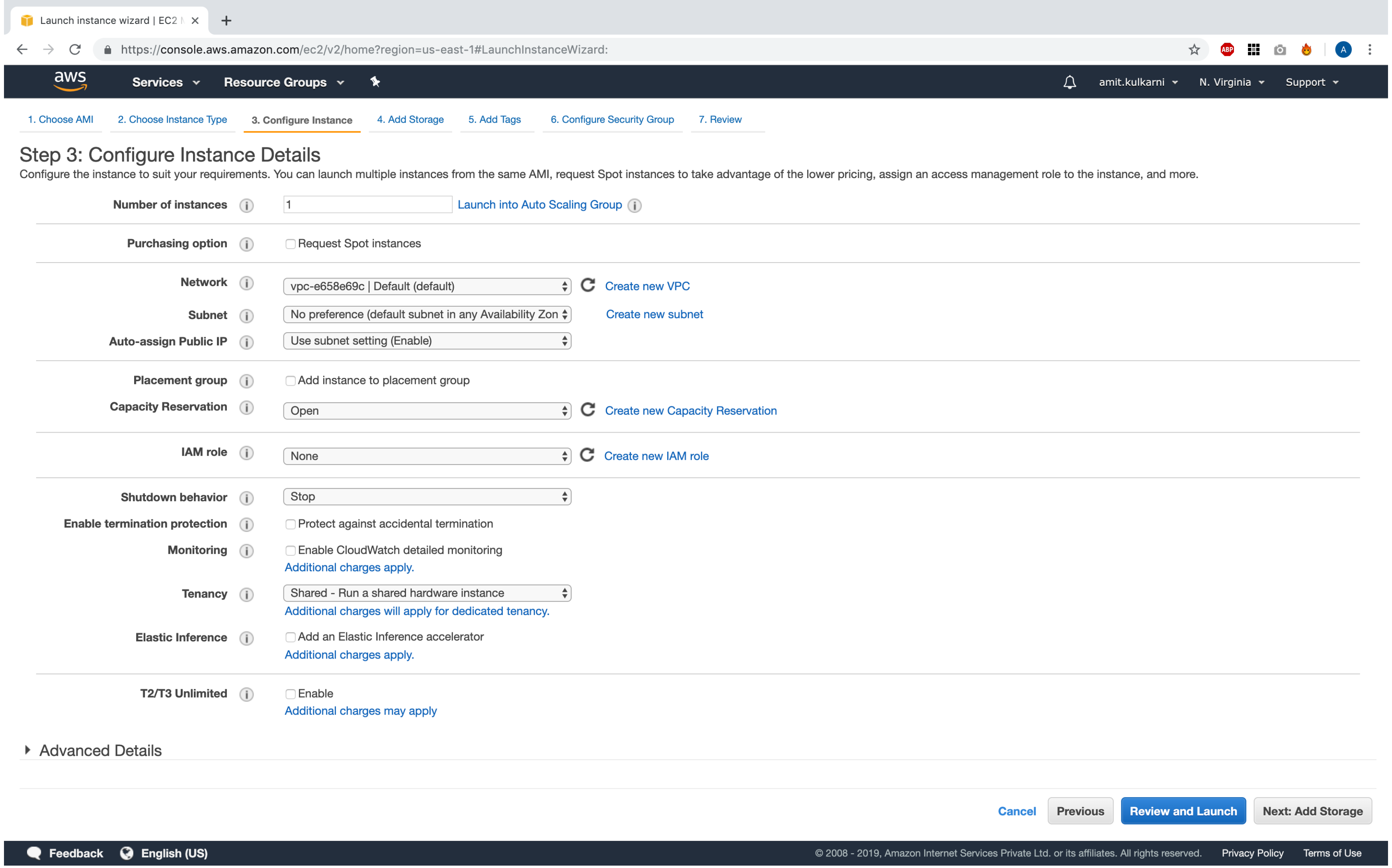
Instance Type

- Used to decide the EC2 instance configuration
- AWS provides various instance types [https://aws.amazon.com/ec2/instance-types/]
  - General purpose: A (ARM), T (Cheapest), M (Main)
  - Compute optimized: C (Compute)
  - Memory optimized: R (RAM), X (Extreme RAM), Z (High compute and memory)
  - Accelerated computing: P (Picture-GPU), G (Graphics), F (Fast)
  - Storage optimized: I (IOPS), D (Data), H (High Disk Throughput)

Instance Types

Type	Category	Description	Use Cases
M5	General Purpose	Balance of compute, memory and network resources	Mid-sized databases
C5	Compute Optimized	Advanced CPUs	Modelling, Analytics
H1	Storage Optimized	Local HDD Storage	Map Reduce
R4	Memory Optimized	More RAM for $	In-memory caching
X1	Memory Optimized	Terabytes of RAM and SSD	In-memory database
I3	IO Optimized	Local SSD storage, high IOPS	NoSQL databases
G3	GPU Graphics	GPUs with video encoders	3d rendering
P3	GPU Compute	GPUs with tensor cores	Machine Learning
F1	Accelerated Computing	FPGA, custom hardware accelerations	Genomics
T2	Burstable	Shared CPUs, lowest cost	Web servers

Step 3 – Configure Instance Details



Purchasing Options

- EC2 instances can be purchased using one of the following methods
    - On Demand
    - Sport instances
    - Reserved instances
    - Dedicated hosts
    - Dedicated instances

On Demand

- Low cost and flexibility with no upfront cost
- Pay for hours used with no commitment
- Good option for developers or testers
- Ideal for
  - auto scaling groups
  - Unpredictable workloads

Spot instances

- Very low hourly compute cost
- Ideal for grid computing and HPC
- Configurable start and end time
- Charged by hour unless AWS terminates in which case that hour will be free
- Can not use encrypted volumes
- Remaining capacity can be sold on AWS market

Reserved instances

- Significant discounts over on demand pricing
- Capacity can be reserved for 1 to 3 years
- Following types of reserved instances can be created
  - Standard
    - Commitment of 1 to 3
    - Charged irrespective of the EC2 instance (whether it is on or off)
  - Convertible
    - Can change the instance family, OS, tenancy and payment options
  - Scheduled
    - Can be reserved for specific periods of time
    - Hourly charges

Dedicated hosts

- AWS uses physical servers for dedicated hosts only for your use
- You can control which instances are deployed on that host
- Available as on demand or with dedicated host reservation
- Good for regulatory compliance or licensing requirements
- Predictable performance as you will be using the physical hardware

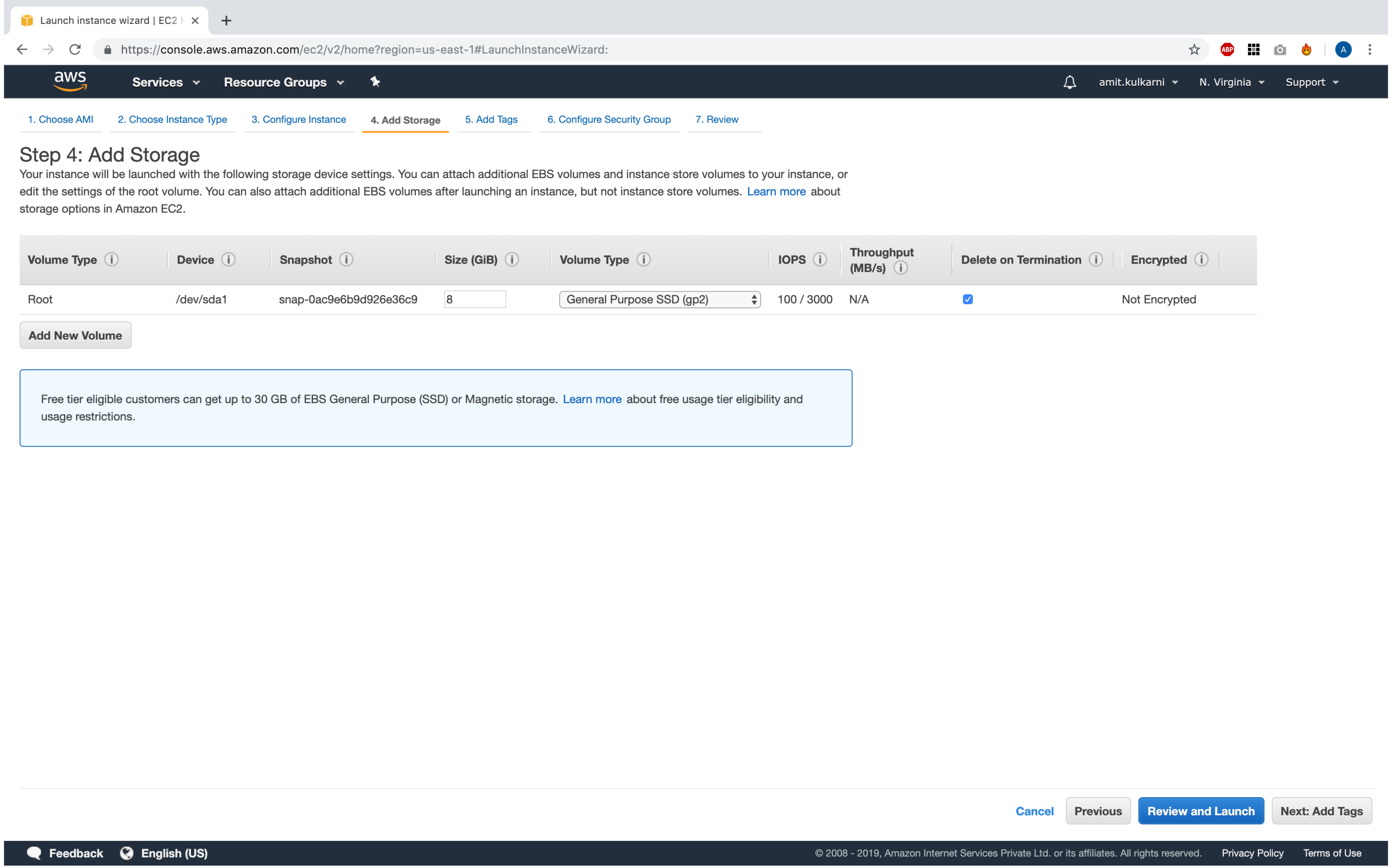
Dedicated instances

- Billing is per instance
- AWS may share the hardware with other non-dedicated instances in the same account
- Available as on demand, reserved instances and spot instances
- Costs additional $2 per hour per region

Placement Groups

- Logical grouping of instances
- Types
  - Cluster
    - Instances are placed in the single availability zone
    - Gives best low latency network performance
  - Spread
    - Instances are spread across underlying hardware across availability zones
    - Reduces risk of simultaneous hardware failure
  - Partition
    - Spreads instances across logical partitions
    - Groups of instances in one partition do not share the underlying hardware with groups of instances in different partitions
    - Generally used in distributed and replicated workloads like Hadoop, Kafka etc

Step 4 – Add Storage



Storage options

- Block storage
  - Instance store
  - EBS volume
- Object storage
  - S3
- File sharing
  - EFS

Instance Store

- Virtual devices whose underlying hardware is physically attached to the host computer that is running the instance
  - Local storage
  - Faster IOPS
- Are ephemeral data (the volumes only exists for duration of life of the instance it is attaching to)
- Once the instance is stopped/shut down, the data is erased
- The instance can be rebooted and still maintain its ephemera data
- Suitable for paging/swap files, caches, replicated data (NoSQL DB)

EBS – Elastic Block Store

- EBS volumes are persistent (they can live beyond the life of the EC2 instance they are attached to)
- Network attached storage (can be attached/detached to or from various EC2 instances)
- Can be attached to only ONE EC2 instance at a time
- Can be backed to S3 in a form of a snapshot which can later be stored into a new EBS volume
- 99.99% available
- The annual failure rate is about 1-2%
- Performance can be measured in IOPS
  - IOPS are input/output per second
  - IOPS measures the number of read and write operations per second, while throughput measures the number of bits read or written per second.
  - AWS measures IOPS in 256K chunk (or smaller)
  - Operations that are greater than 256KB are separated into individual 256KB chunk (512KB operation would count by 2 IOPS)
- Types
  - SSD
  - HDD

EBS – Solid State Drives (SSD)

- General Purpose (gp2)
  - Used for dev/test environments
  - Baseline performance: 100 IOPS – 3000 IOPS
  - Bursts up to 32000 IOPS (credit based)
  - Volume size from 1GiB to 16 TiB
- Provisioned IOPS (io1)
  - Used for mission critical applications that require sustained IOPS performance
  - Large database workloads
  - Volume size: 4GiB to 16TiB
  - Performs at provisioned level and can provision up to 64K IOPS per volume
  - NOTE: EC2 instances have a maximum of 80K IOPS total

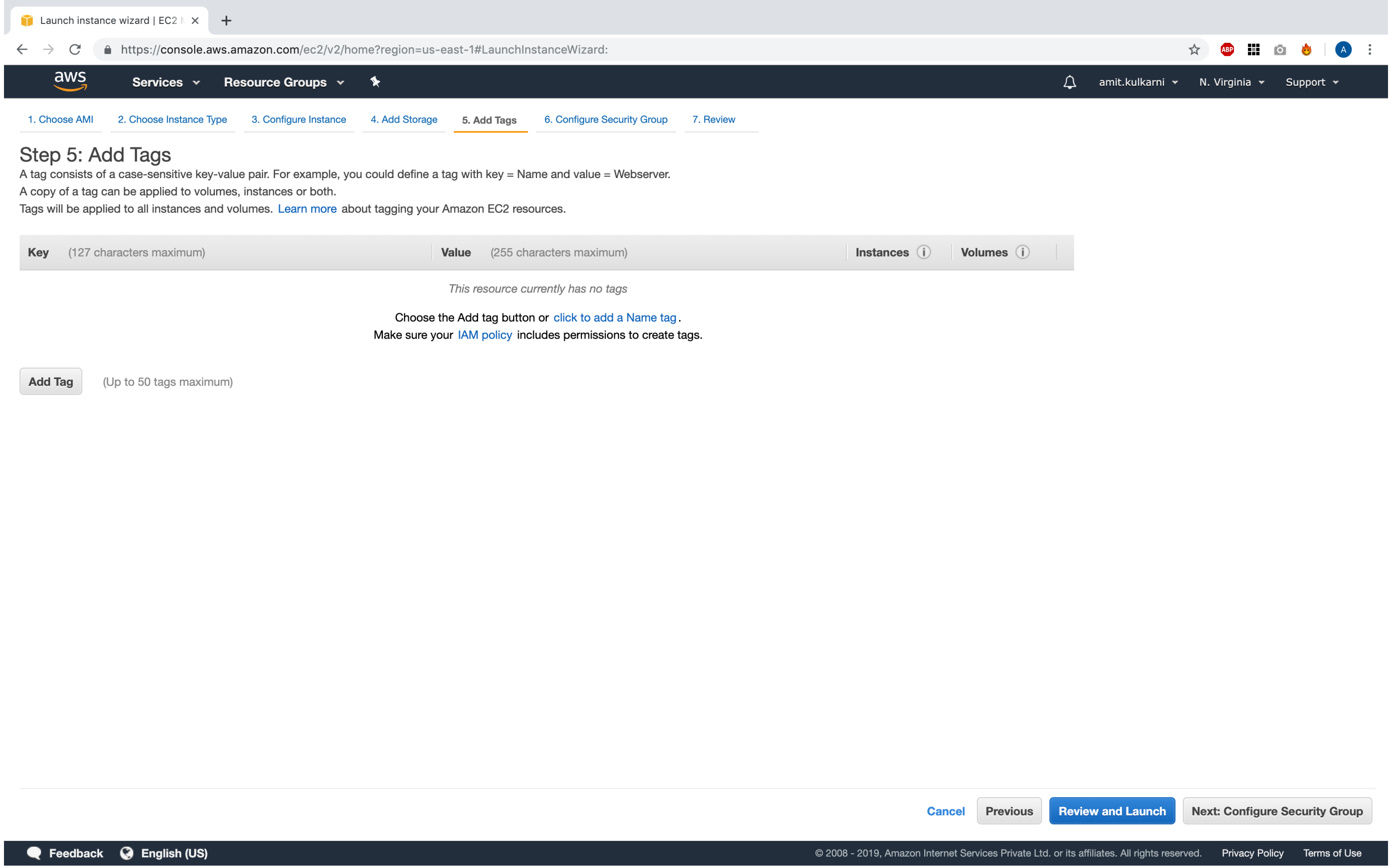
EBS – Hard Disk Drive (HDD)

- Can not be boot volume
- Size: 500GiB – 16TiB
- Types
  - Throughput Optimized (st1)
    - Low storage cost
    - Used for frequently accessed, throughout intensive workloads
    - E.g. Streaming, Big Data
    - Max throughput of 500 MiB/s
    - NOTE: EC2 instances have a maximum throughput of 1750MiB/s
  - Cold HDD (sc1)
    - Lowest cost
    - Infrequently accessed data
    - Max throughput of 250MiB/s

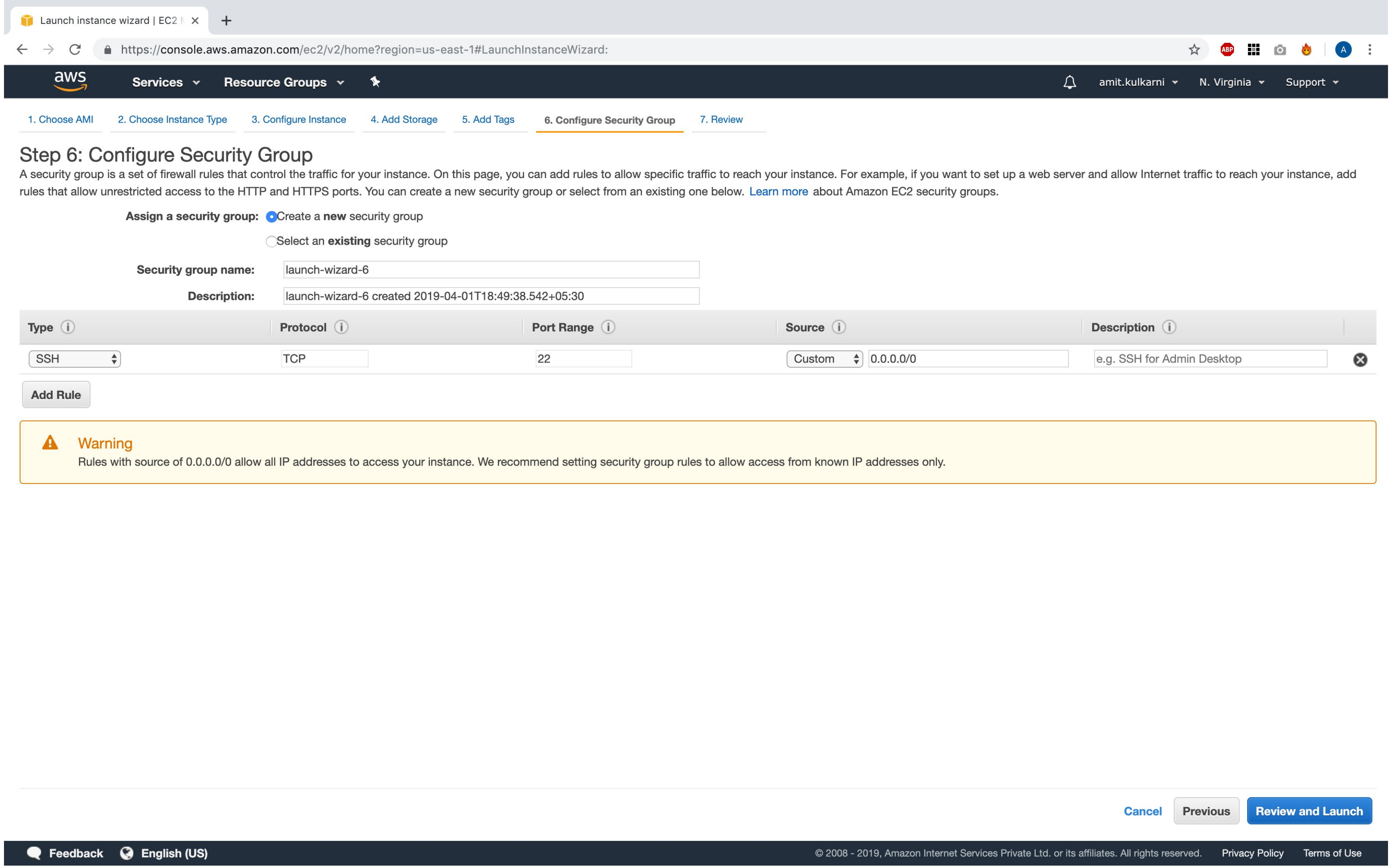
EBS - Snapshots

- Point-in-time backups of EBS volumes that are stored in S3
- They are incremental
- Stores only changes since the most recently taken snapshot, this reducing costs
- If old snapshot is deleted blocks required to restore the other snapshots are retained
- Can be used to create fully restored EBS volumes
- Can be used to create AMIs
- Frequent snapshots of your data increases data durability
- Stop writes before issuing the snapshot command

Step 5 – Add Tags



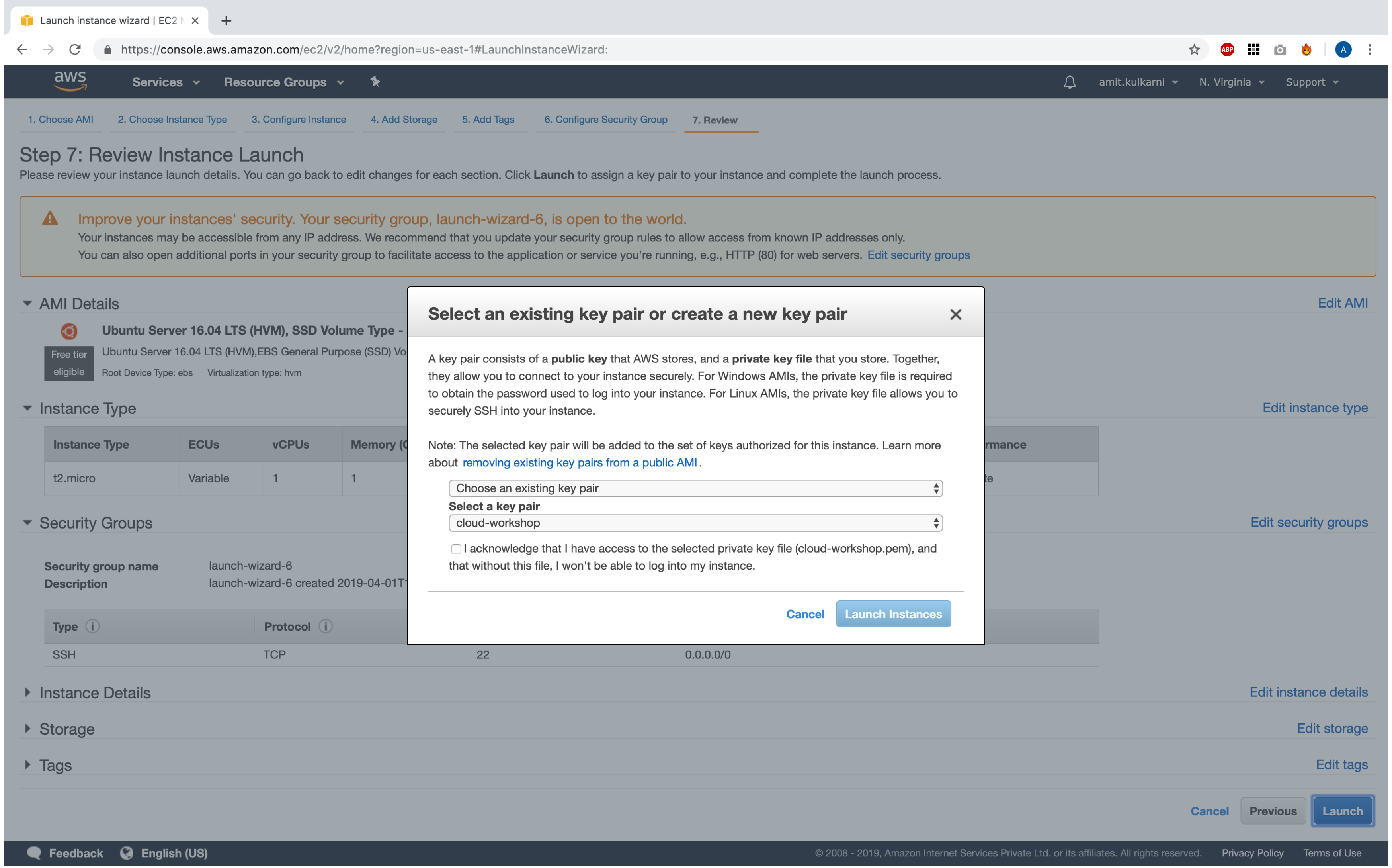
Step 6 - Add Security Group



Security Group

- Acts as a virtual firewall for your instance to control inbound and outbound traffic
- Controls the ports and protocols that can reach the front-end listener
- Every EC2 instance must have at least one security group attached
- Up to 5 security groups can be attached to an EC2 instance
- Security groups act at the instance level, not the subnet level
- Security group contains rules
    - You can specify allow rules, but not deny rules
    - You can specify separate rules for inbound and outbound traffic
    - When you create a security group, it has no inbound rules
    - By default, a security group includes an outbound rule that allows all outbound traffic
    - Security groups are stateful
    - Instances associated with a security group can't talk to each other unless you add rules allowing it
    - Security groups are associated with network interfaces

Step 7 – Review and create PEM file



EC2 Key Pairs

- Uses PEM format (Privacy Enhanced Mail)
- Used to authenticate a client when logging into EC2 instance
- Each key pair consists of a public key and a private key
- AWS stores the public key on the instance and your are responsible for storing the private key
- To log into the instance you must create and authenticate with key pair
  - Linux instances have no password and you use a key pair to log in
  - With windows you use a key pair to obtain the administrator password and then log into the instance with RDP
- During the creation process of an EC2 instance you are required to either create a new key pair or use existing pair
- The private key is available for download and stored on your local drive
- NOTE: it will be available only once in the form of .pem file

S3

Introduction

- Amazon S3 is object storage built to store and retrieve any amount of data from anywhere on the Internet
- It's a simple storage service that offers an extremely durable, highly available, and infinitely scalable data storage infrastructure at very low costs
- Amazon S3 is a distributed architecture and objects are redundantly stored on multiple devices across multiple facilities (AZs) in an Amazon S3 region
- Amazon S3 is a simple key-based object store
- Keys can be any string, and they can be constructed to mimic hierarchical attributes
- Alternatively, you can use S3 Object Tagging to organize your data across all of your S3 buckets and/or prefixes
- Amazon S3 provides a simple, standards-based REST web services interface that is designed to work with any Internet-development toolkit
- Files can be from 0 bytes to 5TB
- The largest object that can be uploaded in a single PUT is 5 gigabytes

Introduction

- For objects larger than 100 megabytes use the Multipart Upload capability
- Updates to an object are atomic - when reading an updated object, you will either get the new object or the old one, you will never get partial or corrupt data
- There is unlimited storage available
- It is recommended to access S3 through SDKs and APIs (the console uses APIs)
- Event notifications for specific actions, can send alerts or trigger actions
- Amazon S3 automatically scales to high request rates
- S3 is a persistent, highly durable data store

Use cases

- **Backup and Storage** – Provide data backup and storage services for others
- **Application Hosting**– Provide services that deploy, install, and manage web applications
- **Media Hosting** – Build a redundant, scalable, and highly available infrastructure that hosts video, photo, or music uploads and downloads
- **Software Delivery** – Host your software applications that customers can download

Charges

- No charge for data transferred between EC2 and S3 in the same region Data transfer into S3 is free of charge
- Data transferred to other regions is charged
- Data Retrieval (applies to S3 Standard-IA and S3 One Zone-IA)
- Charges are:
  - Per GB/month storage fee
  - Data transfer out of S3
  - Upload requests (PUT and GET)
  - Retrieval requests (S3-IA or Glacier)

Multipart upload

- Can be used to speed up uploads to S3
- Multipart upload uploads objects in parts independently, in parallel and in any order
- Performed using the S3 Multipart upload API
- It is recommended for objects of 100MB or larger
    - Can be used for objects from 5MB up to 5TB
    - Must be used for objects larger than 5GB
- If transmission of any part fails it can be retransmitted
- Improves throughput
- Can pause and resume object uploads
- Can begin upload before you know the final object size
- You can create a copy of objects up to 5GB in size in a single atomic operation, but for files larger than 5GB you must use the multipart upload API

S3 Buckets

- Files are stored in buckets:
  - A bucket can be viewed as a container for objects
  - A bucket is a flat container of objects
  - It does not provide a hierarchy of objects
  - You can use an object key name to mimic folders
- 100 buckets per account by default
- You can store unlimited objects in your buckets
- You can create folders in your buckets (only available through the Console)
- You cannot create nested buckets
- Bucket ownership is not transferrable
- Bucket names cannot be changed after they have been created
- If a bucket is deleted its name becomes available again
- Bucket names are part of the URL used to access the bucket

S3 Buckets

- An S3 bucket is region specific
- S3 is a universal namespace so names must be unique globally
- URL is in this format: https://s3-***eu-west-1***.amazonaws.com/***<bucketname>***
- Can backup a bucket to another bucket in another account
- Can enable logging to a bucket
- Bucket naming:
  - Bucket names must be at least 3 and no more than 63 character in length
  - Bucket names must start and end with a lowercase character or a number
  - Bucket names must be a series of one or more labels which are separated by a period
  - Bucket names can contain lowercase letters, numbers and hyphens
  - Bucket names cannot be formatted as an IP address
- For better performance, lower latency, and lower cost, create the bucket closer to your clients

Objects

- Each object is stored and retrieved by a unique key (ID or name)
- An object in S3 is uniquely identified and addressed through:
  - Service end-point
  - Bucket name
  - Object key (name)
  - Optionally, an object version
- Objects stored in a bucket will never leave the region in which they are stored unless you move them to another region or enable cross-region replication
- You can define permissions on objects when uploading and at any time afterwards using the AWS Management Console
- Objects can:
  - Be as small as 0 bytes and as large as 5 TB
  - Have multiple versions (if versioning is enabled)
  - Be made publicly available via a URL
  - Automatically switch to a different storage class or deleted (via lifecycle policies)
  - Encrypted
  - Organized into sub-name spaces called folders

Storage Classes

- There are four S3 storage classes
  - S3 Standard (durable, immediately available, frequently accessed)
  - S3 Standard-IA (durable, immediately available, infrequently accessed)
  - S3 One Zone-IA (lower cost for infrequently accessed data with less resilience)
  - Amazon Glacier (archived data, where you can wait 3-5 hours for access)
- For S3 Standard, S3 Standard-IA, and Amazon Glacier storage classes, your objects are automatically stored across multiple devices spanning a minimum of three Availability Zones
- Objects stored in the S3 One Zone-IA storage class are stored redundantly within a single Availability Zone in the AWS Region you select

Storage Classes

	S3 Standard	S3 Intelligent-Tiering*	S3 Standard-IA	S3 One Zone-IA†	S3 Glacier	S3 Glacier Deep Archive
Designed for durability	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)
Designed for availability	99.99%	99.9%	99.9%	99.5%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99.9%	99.9%
Availability Zones	≥3	≥3	≥3	1	≥3	≥3
Minimum capacity charge per object	N/A	N/A	128KB	128KB	40KB	40KB
Minimum storage duration charge	N/A	30 days	30 days	30 days	90 days	180 days
Retrieval fee	N/A	N/A	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved
First byte latency	milliseconds	millseconds	milliseconds	milliseconds	select minutes or hours	select hours
Storage type	Object	Object	Object	Object	Object	Object
Lifecycle transitions	Yes	Yes	Yes	Yes	Yes	Yes

Static Websites

- S3 can be used to host static websites. Cannot use dynamic content such as PHP, .Net etc
- Automatically scales
- You can use a custom domain name with S3 using a Route 53 Alias record
- When using a custom domain name, the bucket name must be the same as the domain name
- Can enable redirection for the whole domain, pages or specific objects
- URL for the website will be: <bucketname>.s3-website-.amazonaws.com
- Does not support HTTPS/SSL
- Returns an HTML document
- Supports object and bucket level redirects
- Only supports GET and HEAD requests on objects
- Supports publicly readable content only

Pre-Signed URLs

- Pre-signed URLs can be used to provide temporary access to a specific object to those who do not have AWS credentials
- By default all objects are private and can only be accessed by the owner
- To share an object you can either make it public or generate a pre-signed URL Expiration date and time must be configured
- These can be generated using SDKs for Java and .Net and AWS explorer for Visual Studio
- Can be used for downloading and uploading S3 objects

Versioning

- Versioning stores all versions of an object (including all writes and even if an object is deleted)
- Versioning protects against accidental object/data deletion or overwrites
- Enables “roll-back” and “un-delete” capabilities
- Versioning can also be used for data retention and archive
- Old versions count as billable size until they are permanently deleted
- Enabling versioning does not replicate existing objects
- Can be used for backup
- Once enabled versioning cannot be disabled only suspended
- Can be integrated with lifecycle rules

Lifecycle Management

- Used to optimize storage costs, adhere to data retention policies and to keep S3 volumes well-maintained
- Bucket level configuration
- The following actions can be performed:
  - Transition to S3-IA (128Kb and 30 days after creation date)
  - Archive to Glacier (30 days after IA if applicable)
- Objects less than 128KB will not be transitioned to S3 Standard-IA
- Objects must be stored in S3 Standard-IA for at least 30 days
- An object must be in S3 Standard for at least 30 days before it can be transitioned to S3 Standard-IA
- You cannot use a lifecycle policy to move an object from Glacier to S3 Standard or S3 Standard-IA (restore to S3 One Zone-IA and copy)
- Can be applied to current and previous versions

Encryption

- You can securely upload/download your data to Amazon S3 via SSL endpoints using the HTTPS protocol (In Transit - SSL/TLS)

Encryption Option	How it Works
SSE-S3	Use S3's existing encryption key for AES-256
SSE-C	Upload your own AES-256 encryption key which S3 uses when it writes objects
SSE-KMS	Use a key generated and managed by AWS KMS
Client-Side	Encrypt objects using your own local encryption process before uploading to S3

Important S3 Facts

- Objects stay within an AWS region and are synced across all AZ's for extremely high availability (99.99%) and durability (99.999999999%)
- You should always create an S3 bucket in a region that makes sense to its purpose:
  - Serving content to customers
  - Sharing data with EC2
- All regions now support read-after-write consistency for puts of new objects into S3
- Object can be immediately available after putting in S3
- Eventual consistency for puts overwriting existing objects, and deletes of objects used in all regions